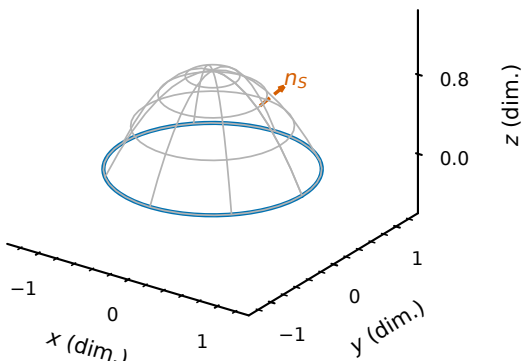
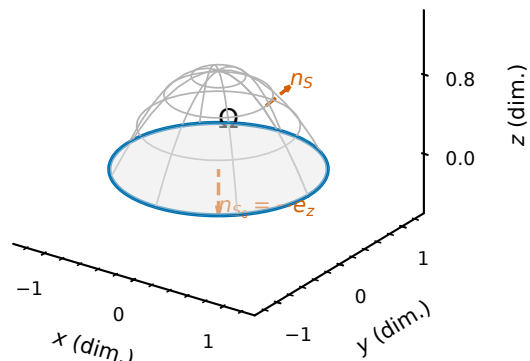
A



S has an edge: $\partial S = C$

B



$S \cup S_0 = \partial\Omega$: closed surface